Actual HW3

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Note: In an effort to avoid "black box" methods, I used Kaiser's `newtraph` for some calculations. However, to properly estimate the MLEs in Q1, some modifications needed to be made to his original function. For that reason, a "modified" `newtraph` function is also attached to this document.

Seho, sorry in advance for the length of this submission (as always).

Q1

Using the mixture model (1) and (2) fit the data in Data Set 1 and Data Set 2 separately using maximum likelihood. Produce approximate 95% confidence intervals for the modes of the distributions – these are the value of the parameter ξ in (2). Conduct any additional analysis needed to construct a likelihood ratio test of a reduced model in which the parameters ξ and θ are identical for both data sets against a full model in which those parameters differ between the two heights. Graph the extreme value density or densities for whichever model (full or reduced) you select.

Data model:

$$f(y \mid \mu_i) = \frac{1}{(2\pi)^{1/2}} \exp\left[-\frac{1}{2}(y-\mu_i)^2\right]. \tag{1}$$

Mixing distribution:

$$g(\mu \mid \xi, \theta) = \frac{1}{\theta} \exp\left(-\frac{\mu-\xi}{\theta}\right) \exp\left[-\exp\left(-\frac{\mu-\xi}{\theta}\right)\right], \qquad \theta > 0. \tag{2}$$

A:

For each height (50m and 80m), the Normal–Extreme Value mixture models were fit by maximum likelihood. Because the latent daily mean μ is unobserved, the likelihood for an observed wind speed y is defined through the marginal mixture density

$$m(y \mid \xi, \theta) = \int f(y \mid \mu)\, g(\mu \mid \xi, \theta)\, d\mu, \qquad \ell(\xi, \theta) = \sum_{i=1}^{n} \log m(y_i \mid \xi, \theta),$$

where f is the Normal density in (1) and g is the Extreme Value density in (2). Each $\ell(\xi, \theta)$ was maximized separately for each data set to obtain MLEs $(\hat{\xi}_{50}, \hat{\theta}_{50})$ and $(\hat{\xi}_{80}, \hat{\theta}_{80})$.

To validate these intervals, profile likelihood intervals were also computed. For fixed ξ (resp. θ), the remaining parameter was re-maximized, yielding the profile log-likelihood. The 95% confidence set consists of all parameter values whose profile log-likelihood is sufficiently close to the maximum, where "sufficiently close" is determined by the likelihood-ratio cutoff:

$$2\{\ell(\hat{\xi}, \hat{\theta}) - \ell(\xi, \hat{\theta}(\xi))\} \leq \chi^2_{1,0.95}.$$

(And an analogous likelihood-ratio cutoff was used when profiling θ.)

Direct Likelihood Evaluation

Table 1: MLEs and Wald 95% confidence intervals for ξ.

Height	$\hat{\xi}$	Lower 95%	Upper 95%
50m	4.392	3.719	5.066
80m	6.717	5.646	7.787

Table 2: MLEs and Wald 95% confidence intervals for θ.

Height	$\hat{\theta}$	Lower 95%	Upper 95%
50m	2.052	1.577	2.671
80m	3.522	2.771	4.477

Profile Likelihood Evaluation

Table 3: Profile likelihood confidence intervals for ξ.

Height	95% Profile Likelihood CI for ξ
50m	[3.700, 5.075]
80m	[5.633, 7.823]

Table 4: Profile likelihood confidence intervals for θ.

Height	95% Profile Likelihood CI for θ
50m	[1.582, 2.686]
80m	[2.793, 4.520]

Likelihood Ratio Test

A likelihood ratio test was constructed comparing a reduced model, in which a common (ξ, θ) was estimated (the reduced-model estimates are omitted, as the full model was ultimately selected) from the pooled data across both heights, to a full model with separate (ξ_{50}, θ_{50}) and (ξ_{80}, θ_{80}).

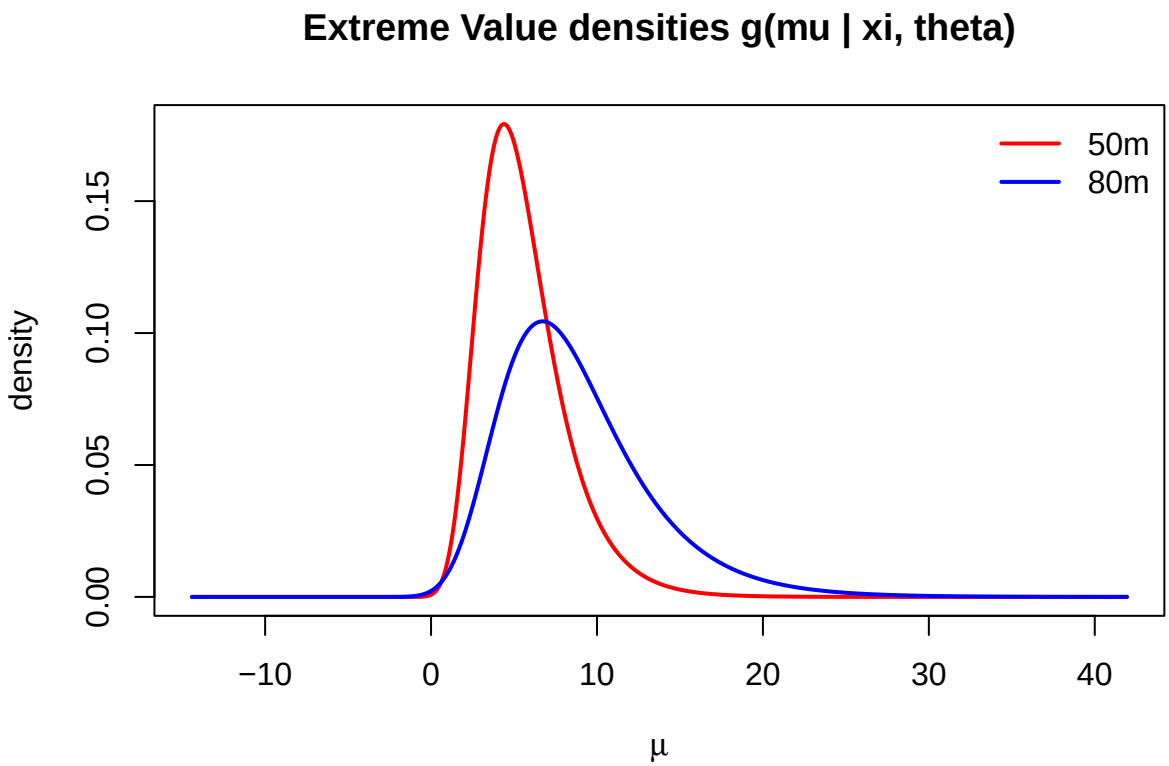
The test statistic

$$\text{LR} = 2\{\ell_{\text{full}} - \ell_{\text{reduced}}\}$$

was compared to a χ^2 distribution with 2 degrees of freedom $(4 - 2)$. The results provided strong evidence against the reduced model, indicating that a common mixing distribution does not adequately describe both heights. Based on this conclusion, the fitted Extreme Value densities $g(\mu \mid \xi, \theta)$ were then plotted under the full model using the corresponding maximum likelihood estimates.

Table 5: Likelihood ratio test for (ξ, θ) (Full vs. Reduced model).

Hypothesis	LR Statistic	df	p-value
H_0: common (ξ, θ) across heights	18.579	2	9.24e-05



Q2

In (2), $E(\mu) \approx \xi + 0.577\theta$, where 0.577 is Euler's constant (don't confuse with Euler's number). Using this expression, produce maximum likelihood estimate or estimates and 95% confidence interval or confidence intervals for the expected true maximum wind speed from your final model. Note that this will also be the expected observed maximum wind speed.

A:

Table 6: MLE and Wald 95% confidence intervals for expected true maximum wind speed.

Height	$\widehat{E(\mu)}$	SE	Lower 95%	Upper 95%
50m	5.577	0.396	4.802	6.353
80m	8.750	0.651	7.474	10.026

Q3

Find a maximum likelihood estimate or estimates (depending on your final model) of the median daily maximum wind speed or speeds. Compute the associated 95% approximate confidence interval or intervals.

A:

The median of Y under the fitted mixture model is defined as the (implicit) solution m to

$$F_Y(m \mid \xi, \theta) = 0.5,$$

where F_Y is the marginal CDF obtained by integrating out the latent μ.

Using the MLEs $(\hat{\xi}, \hat{\theta})$ from Q1, we then solve for

$$F_Y(m \mid \hat{\xi}, \hat{\theta}) = 0.5$$

numerically via root-finding to obtain the estimated median for each height.

Subsequent approximate 95% confidence intervals were then constructed via the delta method, treating the median as a function of $(\xi, \log\theta)$ and using the observed-information covariance from the MLE.

Table 7: MLE and Wald 95% confidence intervals for median daily maximum wind speed.

Height	$\widehat{\text{Median}(Y)}$	SE	Lower 95%	Upper 95%
50m	5.208	0.367	4.488	5.928
80m	8.049	0.601	6.871	9.227

Q4

Find a maximum likelihood estimate or estimates (depending on your final model) for the probability that true daily maximum wind speed is between 4 and 17 m/s. Compute 95% approximate confidence intervals for the estimate or estimates.

A

Under the fitted model, the probability that the latent true wind speed μ lies in $[4, 17]$ is

$$Pr(4 \leq \mu \leq 17) = G(17 \mid \xi, \theta) - G(4 \mid \xi, \theta),$$

where G is the Extreme Value CDF. Using the MLEs $(\hat{\xi}, \hat{\theta})$ from Q1, this expression is evaluated directly to obtain the estimate for each height.

Then, the subsequent approximate 95% confidence intervals were then constructed via the delta method, treating the probability as a function of $(\xi, \log\theta)$ and using the observed-information covariance from the MLE.

Table 8: MLE and 95% Wald confidence intervals for estimated probability $Pr(4 \leq \mu \leq 17)$.

Height	$\widehat{Pr(4 \leq \mu \leq 17)}$	SE	Lower 95%	Upper 95%
50m	0.700	0.060	0.583	0.817
80m	0.832	0.046	0.743	0.922

Q5

What would you conclude about the question of whether there is or is not a benefit to placing turbine hubs at either 50 or 80 m in height?

A:

Overall, there is benefit to placing turbine hubs at 80m.

In Q1, the likelihood ratio test provided evidence that the wind speed distributions at 50m and 80m differ ($p \approx 9.24 \times 10^{-5}$), leading us to adopt the full model with separate parameter estimates for each height. The fitted extreme value mixing distributions therefore represent statistically distinct distributions.

Under the full model, all key summaries indicate systematically higher wind speeds at 80m. The estimated expected true daily maximum wind speed is approximately 3.2 m/s higher at 80m (8.75 vs. 5.58 m/s), and the estimated median is approximately 2.84 m/s higher at 80m (8.05 vs. 5.21 m/s). Further, the estimated probability that true daily maximum wind speed falls within the range of 4–17 m/s is substantially larger at 80m (0.832 vs. 0.700), with confidence intervals suggesting that this difference is unlikely to be explained by sampling variability alone (very minor overlap of probabilities in Q4).

Taken together, a turbine hub height of 80m is associated with more favorable wind speed characteristics, though the above conclusions are conditional on the correctness of the model assumptions and on future wind patterns being similar to those represented in the data.

Q6

As an exercise in learning about statistical procedures, your friend would like to see a Monte Carlo approximation to the expected true wind speed for the two heights separately. Produce such approximations along with 95% Monte Carlo intervals. Your friend understands intuitively that these approximations and the maximum likelihood estimates of Question 2 should have the same or nearly the same values, but marvels at the fact that the intervals associated with these approximations are so much more narrow than those computed using the asymptotic theory of likelihood estimation in Question 2. Explain why this is the case and what it means.

A:

Monte Carlo

Table 9: Monte Carlo approximation of $E(\mu)$ with 95% Monte Carlo intervals.

Height	M	$\widehat{E(\mu)}_{\text{MC}}$	Lower 95%	Upper 95%
50m	20000	5.583	5.546	5.620
80m	20000	8.706	8.644	8.768

Comparison - Concise

The Monte Carlo (MC) intervals are much narrower because they measure the simulation (numerical) error conditional on the estimated parameters $(\hat{\xi}, \hat{\theta})$. By contrast, asymptotic confidence intervals like those in Question 2 account for statistical uncertainty in the joint estimation of (ξ, θ) from finite data.

Comparison - More Details

For each height, we simulated

$$\mu^{(1)}, \ldots, \mu^{(M)} \sim \text{EV}(\hat{\xi}, \hat{\theta})$$

and estimated the expected true wind speed by

$$\widehat{E}_{MC}(\mu) = \bar{\mu}_M = \frac{1}{M} \sum_{m=1}^{M} \mu^{(m)}.$$

By the Central Limit Theorem we know

$$\bar{\mu}_M \approx \mathcal{N}\left(E(\mu), \frac{\text{Var}(\mu)}{M}\right),$$

so an approximate 95% Monte Carlo interval is

$$\bar{\mu}_M \pm 1.96 \frac{s_M}{\sqrt{M}}.$$

The crucial point is that this interval treats $(\hat{\xi}, \hat{\theta})$ as fixed quantities, and is variability in the above formulas is the Monte Carlo approximation error, which decreases at rate $1/\sqrt{M}$ (and can be made arbitrarily small by increasing the number of simulations M).

In contrast, the confidence interval in Question 2 is based on asymptotic likelihood theory and reflects uncertainty in estimating (ξ, θ) from $n = 50$ observations; uncertainty of the parameters propagates into uncertainty about $E(\mu)$.

In toto, the Monte Carlo intervals are narrow because they measure numerical precision and not inferential uncertainty.

Appendix

```r
# Setup
library(tidyverse)
library(numDeriv)
library(here)
source(here("newtraph.txt"))
# Data
m50 <- read.table(here("winddat1.txt"), header = TRUE)
m80 <- read.table(here("winddat2.txt"), header = TRUE)
y50 <- as.numeric(m50$y)
y80 <- as.numeric(m80$y)
# Euler
gamma <- 0.577

# Q1
# Core functions
# input: mu, xi, theta
# output: g(mu | xi, theta) (EV density)
g_mu <- function(mu, xi, theta) {
  z <- (mu - xi) / theta
  (1 / theta) * exp(-z - exp(-z))
}
# input: y, xi, theta; z-bounds for stable integration
# output: m(y | xi, theta) = int f(y|mu) g(mu|xi,theta) dmu
# note: z = (mu - xi)/theta
# m(y) = int phi(y | xi + theta*z, 1) * exp(-z - exp(-z)) dz
m_y_stable <- function(y, xi, theta, zL = -10, zU = 30) {
  integrand <- function(z) {
    log_int <- dnorm(y, mean = xi + theta * z, sd = 1, log = TRUE) - z - exp(-z)
    exp(log_int)
  }
  val <- integrate(
    integrand,
    lower = zL,
    upper = zU,
    rel.tol = 1e-8,
    abs.tol = 0,
    subdivisions = 5000
  )$value
  # numeric stability safeguard
  # one of many
  pmax(val, 1e-250)
}
# input: y vector, xi, theta
# output: vector of log m(y_i | xi, theta)
# convenience function
log_m_vec_stable <- function(y, xi, theta, zL = -10, zU = 30) {
  vapply(X = y, FUN = function(yy) log(m_y_stable(yy, xi, theta, zL, zU)), numeric(1))
}
# input: par = (xi, logtheta), data y
# output: log-likelihood ell(xi,theta)
logLik_one <- function(par, y, zL = -10, zU = 30) {
```

```r
  xi <- par[1]
  theta <- exp(par[2])
  sum(log_m_vec_stable(y, xi, theta, zL = zL, zU = zU))
}
# input: par = (xi, logtheta), data y
# output: negative log-likelihood
# negative log-likelihood because default is min finding
nll_one <- function(par, y, zL = -10, zU = 30) {
  -logLik_one(par, y = y, zL = zL, zU = zU)
}
# input: y vector
# output: start = (xi0, logtheta0)
ev_start <- function(y) {
  v <- var(y)
  theta0 <- sqrt(pmax((v - 1) * 6 / pi^2, 1e-6))
  xi0 <- mean(y) - gamma * theta0
  c(xi0, log(theta0))
}
# input: z bounds
# output: ders(par, dat) -> list(logLik, grad, Hessian)
make_ders_one <- function(zL = -10, zU = 30) {
  function(par, dat) {
    f <- function(pp) logLik_one(pp, y = dat, zL = zL, zU = zU)
    ll <- f(par)
    g <- grad(f, par)
    # hess up, child
    H <- hessian(f, par)
    list(ll, g, H)
  }
}
# input: y vector
# output: MLEs + observed-information covariance on (xi, logtheta) scale
# finally calling an optimization procedure
# feed in good starting values
# feed in safeguarded wrappers
fit_one_newtraph <- function(y, zL = -10, zU = 30) {
  start <- ev_start(y)
  ders <- make_ders_one(zL = zL, zU = zU)
  out <- newtraph(
    ders, dat = y, x0 = start,
    quiet = TRUE, print_final = FALSE,
    step_scale = 0.5, max_halving = 60
  )
  par_hat <- as.numeric(out[[1]])
  xi_hat <- par_hat[1]
  theta_hat <- exp(par_hat[2])
  # inverse hessian
  V <- out[[3]]
  # root information for first entry (xi)
  se_xi <- sqrt(V[1, 1])
  list(
    xi_hat = xi_hat,
    theta_hat = theta_hat,
```

```r
    logLik = out[[2]],
    fInf = V,
    se_xi = se_xi,
    ci_xi_wald = xi_hat + c(-1, 1) * 1.96 * se_xi,
    start = start,
    z_bounds = c(zL, zU)
  )
}
# input: fit from fit_one_newtraph()
# output: Wald CI for theta
# I did this all for xi first, then remembered I needed theta too
# oops
wald_theta_from_fit <- function(fit, alpha = 0.05) {
  z <- qnorm(1 - alpha / 2)

  se_logtheta <- sqrt(fit$fInf[2, 2])
  logtheta_hat <- log(fit$theta_hat)

  ci_logtheta <- logtheta_hat + c(-1, 1) * z * se_logtheta
  se_theta <- fit$theta_hat * se_logtheta

  list(
    theta_hat = fit$theta_hat,
    se_logtheta = se_logtheta,
    se_theta = se_theta,
    ci_theta = exp(ci_logtheta)
  )
}
# MLE
fit50 <- fit_one_newtraph(y50, zL = -100, zU = 100)
fit80 <- fit_one_newtraph(y80, zL = -100, zU = 100)

th50_wald <- wald_theta_from_fit(fit50)
th80_wald <- wald_theta_from_fit(fit80)
# Profile
# input: fixed xi, data y
# output: profile logLik ell_prof(xi)
# note: re-maximize w/r/t theta
profile_logLik_xi <- function(xi_fixed, y, logtheta_center, zL = -10, zU = 30) {
  obj <- function(logtheta) nll_one(c(xi_fixed, logtheta), y = y, zL = zL, zU = zU)

  opt <- optimize(
    obj,
    interval = c(logtheta_center - log(20), logtheta_center + log(20))
  )

  -opt$objective
}
# input: data y and MLEs
# output: CI for xi
ci_xi_profile <- function(y, xi_hat, theta_hat, logLik_hat, zL = -10, zU = 30, alpha = 0.05) {
  cutoff <- qchisq(1 - alpha, df = 1)
```

```r
  target <- function(xi) {
    ell_prof <- profile_logLik_xi(xi_fixed = xi, y = y,
                                  logtheta_center = log(theta_hat), zL = zL, zU = zU)
    2 * (logLik_hat - ell_prof) - cutoff
  }
  step0 <- max(0.25, 0.5 * theta_hat)
  left <- xi_hat - step0
  step <- step0
  for (k in 1:80) {
    tl <- target(left)
    if (is.finite(tl) && tl < 0) { left <- left - step; step <- step * 1.2 } else break
  }
  right <- xi_hat + step0
  step <- step0
  for (k in 1:80) {
    tr <- target(right)
    if (is.finite(tr) && tr < 0) { right <- right + step; step <- step * 1.2 } else break
  }
  # I actually prefer uniroot to bisect
  lower <- uniroot(target, interval = c(left, xi_hat))$root
  upper <- uniroot(target, interval = c(xi_hat, right))$root
  c(lower, upper)
}
# input: fixed logtheta, data y
# output: profile logLik ell_prof(theta)
# note: re-maximize w/r/t xi
profile_logLik_theta <- function(logtheta_fixed, y, xi_center, zL = -10, zU = 30) {
  obj <- function(xi) nll_one(c(xi, logtheta_fixed), y = y, zL = zL, zU = zU)

  theta_fixed <- exp(logtheta_fixed)

  opt <- optimize(
    obj,
    interval = c(xi_center - 20 * theta_fixed, xi_center + 20 * theta_fixed)
  )

  -opt$objective
}
# input: data y and MLEs
# output: LR-based profile CI for theta
ci_theta_profile <- function(y, xi_hat, theta_hat, logLik_hat, zL = -10, zU = 30, alpha = 0.05) {
  cutoff <- qchisq(1 - alpha, df = 1)
  target <- function(logtheta) {
    ell_prof <- profile_logLik_theta(logtheta_fixed = logtheta, y = y,
                                     xi_center = xi_hat, zL = zL, zU = zU)
    2 * (logLik_hat - ell_prof) - cutoff
  }
  lt_hat <- log(theta_hat)
  step0 <- log(1.5)
  left <- lt_hat - step0
  step <- step0
  for (k in 1:80) {
    tl <- target(left)
```

```r
    if (is.finite(tl) && tl < 0) { left <- left - step; step <- step * 1.2 } else break
  }
  right <- lt_hat + step0
  step <- step0
  for (k in 1:80) {
    tr <- target(right)
    if (is.finite(tr) && tr < 0) { right <- right + step; step <- step * 1.2 } else break
  }
  # still prefer uniroot
  lower_lt <- uniroot(target, interval = c(left, lt_hat))$root
  upper_lt <- uniroot(target, interval = c(lt_hat, right))$root
  exp(c(lower_lt, upper_lt))
}
ci50_xi_prof <- ci_xi_profile(y50, fit50$xi_hat,
                              fit50$theta_hat, fit50$logLik,
                              zL = -100, zU = 100)
ci80_xi_prof <- ci_xi_profile(y80, fit80$xi_hat,
                              fit80$theta_hat, fit80$logLik,
                              zL = -100, zU = 100)
ci50_th_prof <- ci_theta_profile(y50, fit50$xi_hat,
                                 fit50$theta_hat, fit50$logLik,
                                 zL = -100, zU = 100)
ci80_th_prof <- ci_theta_profile(y80, fit80$xi_hat,
                                 fit80$theta_hat, fit80$logLik,
                                 zL = -100, zU = 100)
# LRT
# Full
# input: par = (xi50, logth50, xi80, logth80), data y50,y80
# output: ell_full = ell50(xi50,th50) + ell80(xi80,th80)
logLik_full <- function(par, y50, y80, zL = -100, zU = 100) {
  xi50 <- par[1]; th50 <- exp(par[2])
  xi80 <- par[3]; th80 <- exp(par[4])
  sum(log_m_vec_stable(y50, xi50, th50, zL = zL, zU = zU)) +
    sum(log_m_vec_stable(y80, xi80, th80, zL = zL, zU = zU))
}
# Reduced
# input: par = (xi, logtheta), pooled data y50,y80
# output: ell_reduced = ell50(xi,th) + ell80(xi,th)
logLik_reduced <- function(par, y50, y80, zL = -100, zU = 100) {
  xi <- par[1]; th <- exp(par[2])
  sum(log_m_vec_stable(y50, xi, th, zL = zL, zU = zU)) +
    sum(log_m_vec_stable(y80, xi, th, zL = zL, zU = zU))
}
# input: z bounds
# output: list(logLik, grad, Hessian) for full model
# just grabbing the things I need
make_ders_full <- function(zL = -100, zU = 100) {
  function(par, dat) {
    y50 <- dat$y50
    y80 <- dat$y80
    f <- function(pp) logLik_full(pp, y50 = y50, y80 = y80, zL = zL, zU = zU)
    ll <- f(par)
    g <- grad(f, par)
```

```r
    H <- hessian(f, par)
    list(ll, g, H)
  }
}
# input: z bounds
# output: list(logLik, grad, Hessian)
make_ders_reduced <- function(zL = -100, zU = 100) {
  function(par, dat) {
    y50 <- dat$y50
    y80 <- dat$y80
    f <- function(pp) logLik_reduced(pp, y50 = y50, y80 = y80, zL = zL, zU = zU)
    ll <- f(par)
    g <- grad(f, par)
    H <- hessian(f, par)
    list(ll, g, H)
  }
}
# input: start = (xi50, logth50, xi80, logth80)
# output: full-model MLE + logLik
fit_full_newtraph <- function(y50, y80, start, zL = -100, zU = 100) {
  ders <- make_ders_full(zL = zL, zU = zU)
  out <- newtraph(
    ders, dat = list(y50 = y50, y80 = y80), x0 = start,
    quiet = TRUE, print_final = FALSE,
    step_scale = 0.5, max_halving = 60
  )
  par_hat <- as.numeric(out[[1]])
  list(
    xi50_hat = par_hat[1], th50_hat = exp(par_hat[2]),
    xi80_hat = par_hat[3], th80_hat = exp(par_hat[4]),
    logLik = out[[2]],
    fInf = out[[3]],
    start = start,
    out = out
  )
}
# input: start = (xi, logtheta)
# output: reduced-model MLE + logLik
fit_reduced_newtraph <- function(y50, y80, start, zL = -100, zU = 100) {
  ders <- make_ders_reduced(zL = zL, zU = zU)
  out <- newtraph(
    ders, dat = list(y50 = y50, y80 = y80), x0 = start,
    quiet = TRUE, print_final = FALSE,
    step_scale = 0.5, max_halving = 60
  )
  par_hat <- as.numeric(out[[1]])
  list(
    xi_hat = par_hat[1], th_hat = exp(par_hat[2]),
    logLik = out[[2]],
    fInf = out[[3]],
    start = start,
    out = out
  )
```

```r
}
# Actual LRT calculations
start_full <- c(fit50$xi_hat, log(fit50$theta_hat), fit80$xi_hat, log(fit80$theta_hat))
start_red  <- ev_start(c(y50, y80))
full_hat <- fit_full_newtraph(y50, y80, start = start_full, zL = -100, zU = 100)
red_hat  <- fit_reduced_newtraph(y50, y80, start = start_red,  zL = -100, zU = 100)
LR <- 2 * (full_hat$logLik - red_hat$logLik)
pval <- 1 - pchisq(LR, df = 2)
# Plot Work
mu_grid <- seq(
  min(full$xi50_hat - 6*full$th50_hat, full$xi80_hat - 6*full$th80_hat),
  max(full$xi50_hat + 10*full$th50_hat, full$xi80_hat + 10*full$th80_hat),
  length.out = 800
)
# CDF function (given)
g_mu <- function(mu, xi, theta) {
  z <- (mu - xi) / theta
  (1/theta) * exp(-z - exp(-z))
}
plot(mu_grid,
     g_mu(mu_grid, full$xi50_hat, full$th50_hat),
     type = "l",
     lwd = 2,
     col = "red",
     xlab = expression(mu),
     ylab = "density",
     main = "Extreme Value densities g(mu | xi, theta)")
lines(mu_grid,
      g_mu(mu_grid, full$xi80_hat, full$th80_hat),
      lwd = 2,
      col = "blue")
legend("topright",
       legend = c("50m", "80m"),
       col = c("red", "blue"),
       lwd = 2,
       bty = "n")

# Q2

# input: fit with (xi_hat, theta_hat, fInf) on (xi, logtheta) scale
# output: eta_hat, SE, Wald CI for E(mu)
wald_ci_mean_mu <- function(fit) {
  xi_hat <- fit$xi_hat
  th_hat <- fit$theta_hat
  V <- fit$fInf
  eta_hat <- xi_hat + gamma * th_hat
  # d/dxi = 1
  # d/dlogtheta = gamma * theta
  grad_vec <- c(1, gamma * th_hat)
  se <- sqrt(drop(t(grad_vec) %*% V %*% grad_vec))
  ci <- eta_hat + c(-1, 1) * qnorm(0.975) * se
  list(eta_hat = eta_hat, se = se, ci95 = ci)
}
```

```r
ci50_eta <- wald_ci_mean_mu(fit50)
ci80_eta <- wald_ci_mean_mu(fit80)

# Q3
# input: y, xi, theta
# output: F_Y(y | xi, theta)
FY_stable <- function(y, xi, theta, zL = -100, zU = 100) {
  integrand <- function(z) {
    pnorm(y - (xi + theta * z)) * exp(-z - exp(-z))
  }
  val <- integrate(integrand, lower = zL, upper = zU,
                   rel.tol = 1e-8, abs.tol = 0, subdivisions = 5000)$value
  pmin(pmax(val, 0), 1)
}
# input: xi, theta
# output: m solving F_Y(m | xi,theta) = 0.5
median_y <- function(xi, theta, zL = -100, zU = 100) {
  f <- function(y) FY_stable(y, xi, theta, zL, zU) - 0.5

  y0 <- xi + gamma * theta
  lo <- y0 - 10 * theta - 5
  hi <- y0 + 10 * theta + 5
  for (k in 1:60) if (f(lo) > 0) lo <- lo - (2 + theta) else break
  for (k in 1:60) if (f(hi) < 0) hi <- hi + (2 + theta) else break
  uniroot(f, interval = c(lo, hi))$root
}
# input: fit on (xi, logtheta) scale
# output: median plug-in + delta-method Wald CI
wald_ci_median_y <- function(fit, zL = -100, zU = 100) {
  xi_hat <- fit$xi_hat
  th_hat <- fit$theta_hat
  V <- fit$fInf
  h <- function(par) {
    xi <- par[1]
    th <- exp(par[2])
    median_y(xi, th, zL = zL, zU = zU)
  }
  par_hat <- c(xi_hat, log(th_hat))
  m_hat <- h(par_hat)
  g <- grad(h, par_hat)
  se <- sqrt(drop(t(g) %*% V %*% g))
  ci <- m_hat + c(-1, 1) * qnorm(0.975) * se
  list(median_hat = m_hat, se = se, ci95 = ci)
}
ci50_med <- wald_ci_median_y(fit50, zL = -100, zU = 100)
ci80_med <- wald_ci_median_y(fit80, zL = -100, zU = 100)

# Q4
# input: u, xi, theta
# output: EV CDF G(u | xi,theta)
G_mu <- function(u, xi, theta) {
  exp(-exp(-(u - xi) / theta))
}
```

```r
# input: fit on (xi, logtheta) scale
# output: plug-in probability + Wald CI
wald_ci_p_between_mu <- function(fit, a = 4, b = 17) {
  xi_hat <- fit$xi_hat
  th_hat <- fit$theta_hat
  V <- fit$fInf
  h <- function(par) {
    xi <- par[1]
    th <- exp(par[2])
    G_mu(b, xi, th) - G_mu(a, xi, th)
  }
  par_hat <- c(xi_hat, log(th_hat))
  p_hat <- h(par_hat)
  g <- grad(h, par_hat)
  se <- sqrt(drop(t(g) %*% V %*% g))
  ci <- p_hat + c(-1, 1) * qnorm(0.975) * se
  ci <- pmin(pmax(ci, 0), 1)
  list(p_hat = p_hat, se = se, ci95 = ci)
}
ci50_p <- wald_ci_p_between_mu(fit50, a = 4, b = 17)
ci80_p <- wald_ci_p_between_mu(fit80, a = 4, b = 17)

# Q5
# Words

# Q6
set.seed(43)
# input: n, xi, theta
# output: n draws from EV(xi,theta) via inverse-CDF
rEV <- function(n, xi, theta) {
  u <- runif(n)
  xi - theta * log(-log(u))
}
# input: fit with (xi_hat, theta_hat); number of draws M
# output: MC mean + MC CI; also plug-in xi + gamma*theta for comparison
mc_E_mu <- function(fit, M = 20000) {
  xi_hat <- fit$xi_hat
  th_hat <- fit$theta_hat
  mu_draws <- rEV(M, xi_hat, th_hat)
  mc_mean <- mean(mu_draws)
  mc_se <- sd(mu_draws) / sqrt(M)
  mc_ci <- mc_mean + c(-1, 1) * qnorm(0.975) * mc_se
  plug_in <- xi_hat + gamma * th_hat
  list(M = M, plug_in = plug_in, mc_mean = mc_mean, mc_ci95 = mc_ci)
}
mc50 <- mc_E_mu(fit50, M = 20000)
mc80 <- mc_E_mu(fit80, M = 20000)
```